

DataLink and Product Streamer

1 GB Download Benchmark

5 October 2026 | Download start: 08:10:54 UTC (16:10:54 China time)
Endpoint: http://64.176.188.89:18080 | **Product:** lab.test:large-1tb.fits

Test configuration

The Python client first measured a 10-second, single-stream reverse TCP iperf3 baseline from the temporary VM to the client. It then resolved the product through DataLink and downloaded the first **1 GB (1,000,000,000 bytes)** through Product Streamer, with no total download time limit. The source was a sparse, zero-filled FITS file of approximately 1 TB. Received blocks were checked and discarded.

Results

Measurement	Result
iperf3 receiver throughput	22.327 Mbps / 2.662 MiB/s
iperf3 duration / received bytes	10.003 s / 27,918,336 bytes
Streamer download size	1,000,000,000 bytes (1 GB)
Download duration	301.038 s (5 min 1 sec)
Download average throughput	26.575 Mbps / 3.168 MiB/s
Time to first byte	0.608 s
Download / iperf3 throughput	119.03%
Completion / sample content check	PASS / PASS

Data integrity verification

The client checked the HTTP status, Content-Length, Content-Range and absence of compression, and confirmed receipt of all requested bytes. It checked the FITS header's **SIMPLE** and **END** markers and verified that **every received byte after the 2,880-byte header was zero**, matching the synthetic payload. These checks passed for the complete 1 GB sample.

SHA-256 was not computed or compared with an independent reference digest. The header check covered these markers rather than full FITS validation. Verification applies to the downloaded sample, not the entire 1 TB source. A subsequent integrity test should compare a streaming SHA-256 with a separately prepared digest for the exact requested range.

Assessment

The 1 GB sample completed at an average **26.575 Mbps**. Its rate exceeded the preceding iperf3 result. This can occur when sequential tests experience different network conditions, TCP startup effects or port policies; the cause was not established here. The differing durations also affect comparability. Repeat paired measurements to assess variability. This single comparison does not establish a maximum network rate or identify a Streamer bottleneck. A complete 1 TB transfer and CNSRC-node performance were not tested.

Evidence: User-supplied JSON result; no additional transfer was performed to prepare this report.